

Cislunar Orbit Determination

Implementing Extended Kalman Filter and Batch Least squares for estimation of cislunar trajectories

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Research Projects

Author Note

Documentation of relevant projects and assignments related to spacecraft GN&C algorithm analysis

Abstract

Basic implementation of relevant GN&C algorithms is shown, including Extended Kalman Filter (EKF) for sequential estimation and Batch Least Squares for initial orbit determination. To model cislunar dynamics, the Circular Restricted Three Body Problem (CRTBP) equations of motion are used to propagate the states of a spacecraft within the Earth-Moon system. Using a range/range-rate measurement model, noisy measurements are generated to test each algorithm's convergence. The results are Monte Carlo simulations performed to evaluate robustness against uncertain conditions such as process or sensor noise.

Keywords: Estimation, EKF, Batch Least Squares, Monte Carlo Simulations

System and Measurement Model

The Circular Restricted Three Body Problem (CRTBP) is the dynamic model used to evaluate the state derivatives, $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x})$, and propagate the spacecraft's trajectory over time. This is a nonlinear system that must be integrated numerically. In this model, the states are defined with respect to a rotating frame with angular velocity such that the Earth and Moon are fixed along the x-axis. The cislunar orbit that is used to simulate the spacecraft's states, $\mathbf{x}(t) = \begin{bmatrix} \mathbf{r}_{sc} \\ \mathbf{v}_{sc} \end{bmatrix}$, is shown in Figure 1 relative to the moon's center. It is an unstable northern L_1 Halo orbit with a period of approximately 12 days with given initial conditions that start at perilune.

The spacecraft's uncertainties are represented in the state covariance matrix, $P(t)$, which are propagated using Lyapunov's matrix differential equations:

$$\dot{P}(t) = F(\mathbf{x}(t))P(t) + P(t)F^T(\mathbf{x}(t)) + G(t)Q(t)G^T(t)$$

$$F(t) = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right]_{\mathbf{x}(t)}$$

The matrix $G(t)$ is the mapping of process noise to the state derivatives and $Q(t)$ is the covariance associated with this random variable. For this implementation, however, this noise is not included.

To estimate the spacecraft's state, radar sensors are assumed to provide range and range-rate measurements, $\mathbf{y} = \begin{bmatrix} \rho \\ \dot{\rho} \end{bmatrix}$, with respect to an observer at an earth-fixed location defined by longitude and latitude. The observer's position and velocity ($\mathbf{r}_{ob}$ and $\mathbf{v}_{ob}$) are propagated forward at each tracking time accounting for the speed and direction of the Earth's spin. The measurement model, $\mathbf{h}(\mathbf{x}, t)$, is a nonlinear function of the spacecraft's states relative to the observer and is obtained by the following equations.

$$range: \rho = \sqrt{\boldsymbol{\rho}^T \boldsymbol{\rho}} \quad range - rate: \dot{\rho} = \frac{1}{\rho} \dot{\boldsymbol{\rho}}^T \boldsymbol{\rho}$$

$$, \text{ where relative position: } \boldsymbol{\rho} = \mathbf{r}_{sc} - \mathbf{r}_{ob} \text{ and relative velocity: } \dot{\boldsymbol{\rho}} = \mathbf{v}_{sc} - \mathbf{v}_{ob}$$

For this simulation, an arbitrary location of $(long, lat) = (0, 45^\circ)$ is chosen. The range and range-rate standard deviations are defined to simulate sensor noise:

Measurement	Standard deviation (σ)
Range	5 m
Range-rate	5 mm/s

For each simulation, a new set of noisy measurements are generated based on normally distributed random errors with standard deviations defined by the measurement covariance. The tracking times where the measurements are taken were chosen to be a set of 60 measurements equally spaced in time over the 12-day period. The simulation requires an initial guess that can be arbitrarily chosen or fed through another method. In this case, the initial state is centered at the true initial state with initial covariance was defined as:

State variable	Standard deviation (σ)
Position	100 m
Range-rate	1 cm/s

Extended Kalman Filter

For sequential estimation of orbital states, Kalman Filters are used to update the current state in real time as new measurements become available. The spacecraft's states and variances are propagated forward in time to obtain a predicted estimate, $\hat{\mathbf{x}}_i^-$, and expected measurement, $\mathbf{h}(\hat{\mathbf{x}}_i^-)$. The estimate is then corrected given the actual measurement, $\mathbf{y}_i$, as follows.

$$\hat{\mathbf{x}}_i^+ = \hat{\mathbf{x}}_i^- + K_i(\mathbf{y}_i - \mathbf{h}(\hat{\mathbf{x}}_i^-))$$

$$P_i^+ = (I - K_i H_i) P_i^-$$

The Kalman gain, K_i , is a function of the propagated state covariance matrix, P_i^- , and the sensor covariance matrix, R .

$$K_i = P_i^- H_i^T (H_i P_i^- H_i^T + R)^{-1}$$

The Extended Kalman Filter (EKF) is implemented due to the nonlinear dynamics and measurement model, which requires the linearization of these functions and evaluation at each current state estimate. Therefore, the matrix H_i becomes the partial derivative of the measurement equations with respect to the spacecraft's states and is evaluated at the current estimate before the update.

$$H_i = \left[\frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right]_{\hat{\mathbf{x}}_i^-}^{2 \times 6} = \left[\frac{\partial \rho}{\partial \mathbf{p}} \quad \frac{\partial \rho}{\partial \dot{\mathbf{p}}} \right]_{\hat{\mathbf{x}}_i^-} = \frac{1}{\rho} \begin{bmatrix} \boldsymbol{\rho}^T & 0^{1 \times 3} \\ (1 - \frac{\boldsymbol{\rho}^T \dot{\boldsymbol{\rho}}}{\rho^2}) \boldsymbol{\rho}^T & \boldsymbol{\rho}^T \end{bmatrix}_{\hat{\mathbf{x}}_i^-}$$

Batch Least Square Estimation

Batch Least Squares (BLS) is used for initial orbit determination to estimate the initial conditions based on a set of noisy measurements. In this method, however, the measurements are used simultaneously in one batch, to correct the initial state estimate. This algorithm is derived such that the sum of squared error between expected and actual measurements, $\Delta y_i = y_i - \mathbf{h}(\hat{\mathbf{x}}_i)$ is minimized.

$$J_i = \Delta \mathbf{y}_c^T W \Delta \mathbf{y}_c$$

$$\Delta \mathbf{y}_c = \begin{bmatrix} \Delta y_0 \\ \vdots \\ \Delta y_{n-1} \\ \Delta y_n \end{bmatrix}$$

To handle the nonlinear dynamics, the estimate is obtained iteratively until the difference between corrections is negligible with some tolerance, ϵ .

$$\mathbf{x}_{new} = \mathbf{x}_c + \Delta \mathbf{x}_c$$

$$\Delta \mathbf{x}_c = (H^T W H)^{-1} H^T W \Delta \mathbf{y}_c$$

$$W = R^{-1} \text{ and } H = \begin{bmatrix} \frac{\partial h(\hat{\mathbf{x}}_0)}{\partial \mathbf{x}_0} \\ \vdots \\ \frac{\partial h(\hat{\mathbf{x}}_{n-1})}{\partial \mathbf{x}_0} \\ \frac{\partial h(\hat{\mathbf{x}}_n)}{\partial \mathbf{x}_0} \end{bmatrix}$$

The block matrix H is composed of partial derivatives of measurement model with respect to the initial state evaluated at each of the track times that the measurements are taken. These can be evaluated as follows:

$$\frac{\partial h(\hat{\mathbf{x}}_i)}{\partial \mathbf{x}_0} = \left[\frac{\partial h}{\partial \mathbf{x}} \right]_{\hat{\mathbf{x}}_i} \frac{\partial \mathbf{x}_i}{\partial \mathbf{x}_0} = H_i \Phi(t_0, t_i)$$

The state transition matrix, $\Phi(t)$, is obtained by propagating the equations:

$$\dot{\Phi}(t) = A(t)\Phi(t), \text{ and } \Phi(t_0, t_0) = I^{6 \times 6}$$

Finally, the initial covariance associated with the resulting estimate is obtained from the converged values.

$$P_0 = (H^T W H)^{-1} \text{ and } P(t) = \Phi(t_0, t) P_0 \Phi^T(t_0, t)$$

Monte Carlo Results and Analysis

The results for a Monte Carlo simulation of only 100 runs are shown for both estimation methods in Figures 2 and 3. The position and velocity errors are compared against the integrated values. The state uncertainty is outlined through the 3σ values, obtained from the average values from all the covariance results. These show the bounds where $\sim 90\%$ of state estimates are expected to fall. Every instance of the simulation falls within these confidence bounds, showing calculation of state estimates and uncertainties is consistent.

For the EKF results, shown in Figure 2, the uncertainty propagation is proved to converge to lower values of position and velocity errors over time for all simulations, showing that subsequent measurements improve confidence in the state estimate. The transient behavior before the filter converges to smaller errors can be explained by the variation in geometry of the spacecraft's trajectory relative to the observer. The jumps at the time of each measurement show that the uncertainties effectively decrease after the updated estimate is calculated. Meanwhile the predicted estimates that are propagated between measurements do not have a particular trend. In general, they may increase due to the unstable nature of these Halo orbits.

In contrast, the covariance results for BLS, in Figure 3, don't show the same trend because they are propagated only from the initial covariance and change with the flow of variational dynamics. Their profile show the nature of the unstable dynamics relative to the halo orbit. The biggest variation, for example, is seen during the middle of the propagation near apolune.

The error standard deviations over time are also compared against the 1σ values at the bottom of these Figures. These show that for both methods, they fairly match the expected standard deviations obtained from the covariance results. It is expected that these values will fall more closely as the number of Monte Carlo runs is increased.

As a final analysis, the number of measurements is decreased to compared the variation in the results. In Figure 4, the average standard deviations are again compared to the error standard deviations from EKF and BLS results using 30 measurements over Monte Carlo 100 runs. As expected from the EKF (top figure), the errors and standard deviations are allowed to grow larger than in the original results and they seem to converge at longer times. The BLS results also show convergence at larger errors in the initial and propagated states.

Conclusions

Both methods are successful at converging to state estimates that account for the sensor noise present under real world conditions. The Monte Carlo simulations show robustness against these uncertainties and converge to close estimates. Variation of parameters such as number of tracks/measurements is analyzed and results from both estimation methods behave as expected, showing better convergence and smaller uncertainties with a larger number of measurements.

References

Crassidis, J. L., & Junkins, J. L. (2004). *Optimal estimation of dynamic systems*. Chapman and Hall/CRC. Pages 188 and 407.

Figures

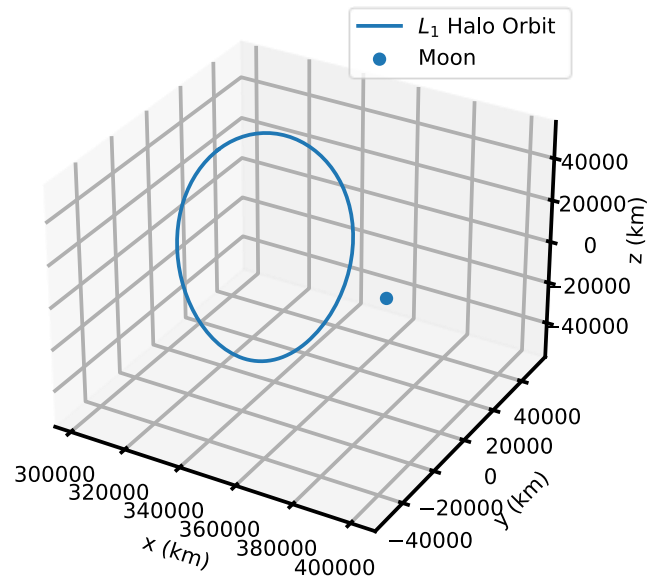
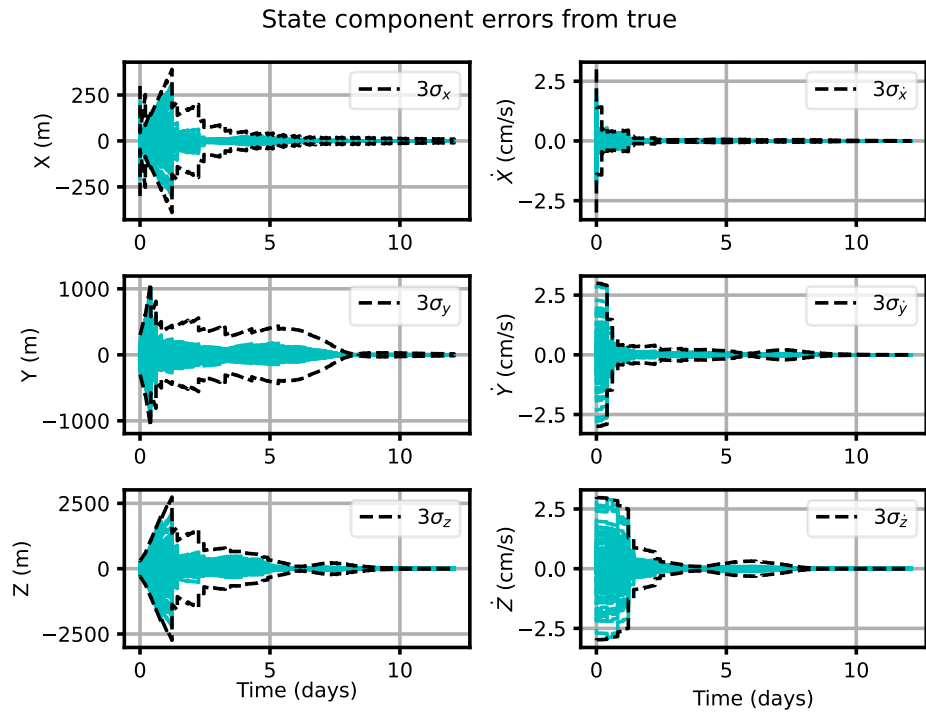
Figure 1. L_1 Halo orbit visualization with the moon as reference

Figure 2. Monte Carlo results for EKF simulation using 60 measurements



State component error std's vs average std's from true (Monte carlo)

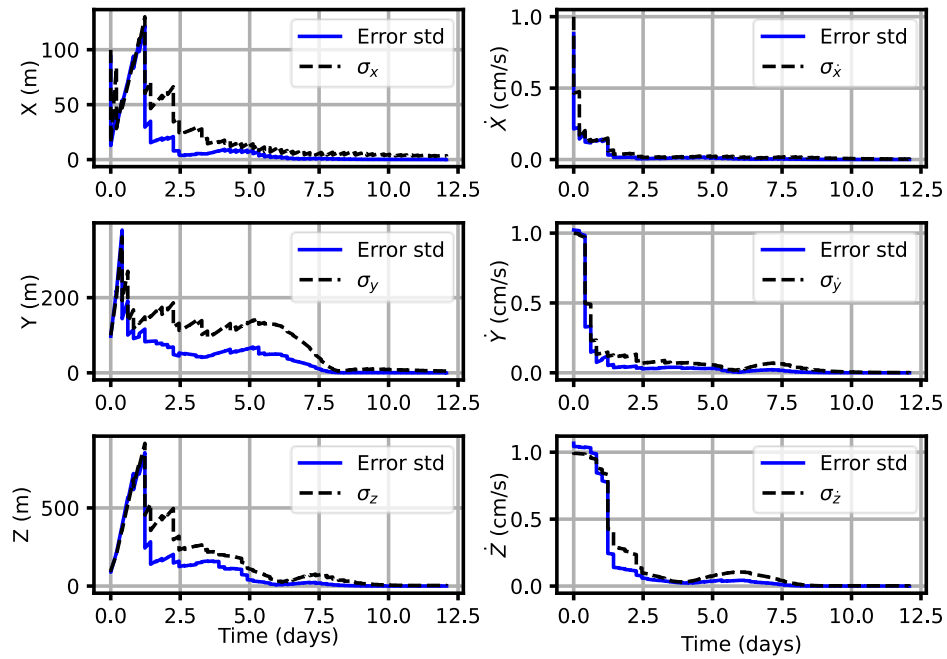


Figure 3. Monte Carlo results for BLS simulation using 60 measurements

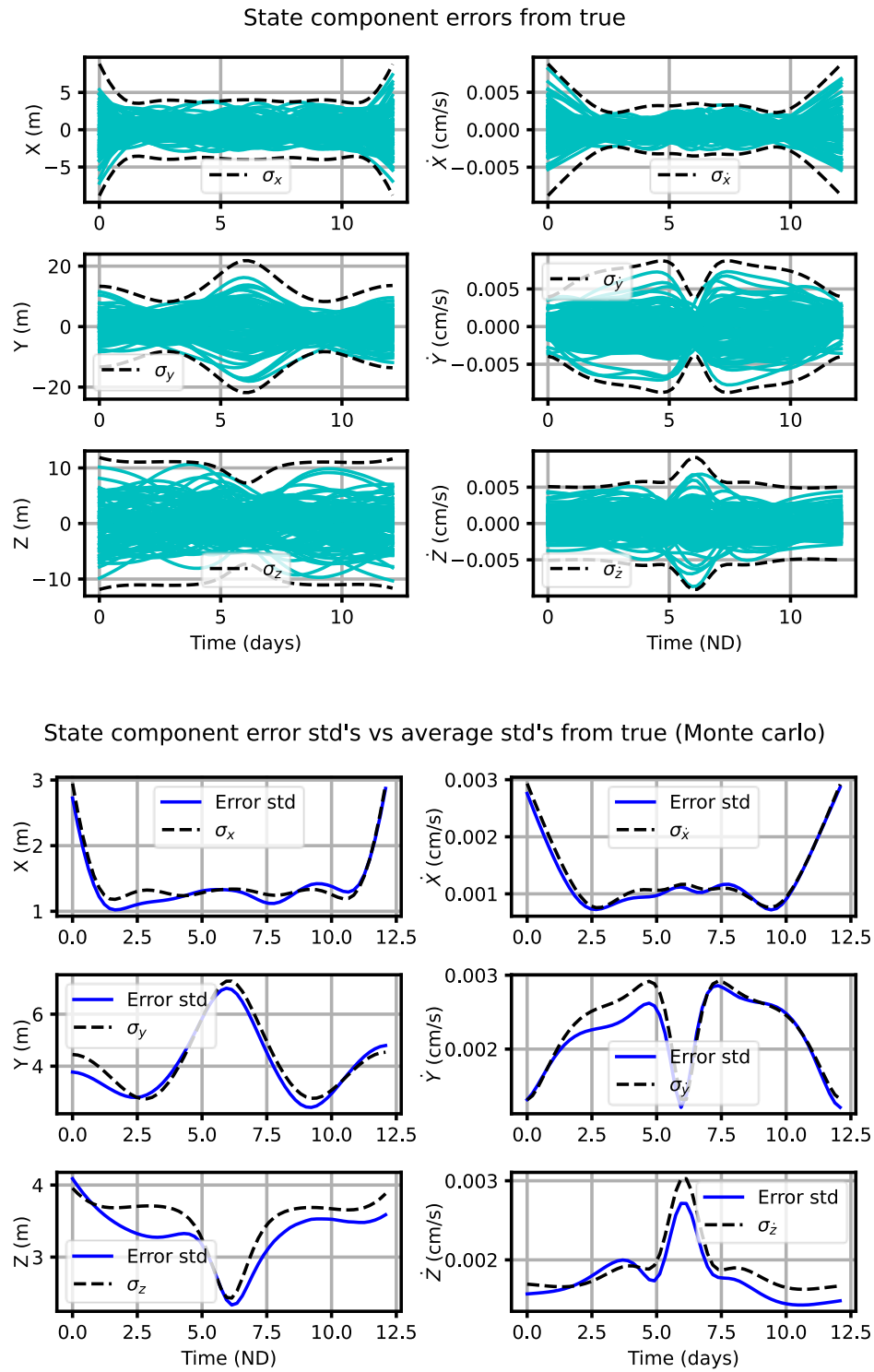


Figure 4. Average Monte Carlo results for EKF and BLS simulation using 30 measurements

